



Vishal Sharma

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Professional Summary

A machine learning engineer passionate about cutting-edge technology and solving real-world problems, with experience in applied **Natural Language Processing and Generative AI in the HR-Tech and Fin-Tech domain.**

Skills & Research

Key Skills:

- Machine Learning | Deep Learning
- NLP | NLU | NLG | Generative AI
- LLMs | Vector search | LangChain
- Prompt Engineering | Content Creation

Technical Skills:

Machine Learning & Deep Learning

- Supervised & Unsupervised Algorithms
- Text Processing & Embeddings like TFIDF, Word2Vec, Glove, and BERT
- Neural Network Architectures for text like RNN, LSTM, GRU, Seq-2-Seq, ULFMT, transformers, encoder-decoder and T5 model
- Open source LLMs, Vector-DB (Qdrant), searching & Ranking, OpenAI models
- Open source STT, TTS, and APIs

Languages & Scripting:

Python, C (Basic), SQL, HTML & CSS (Basic)

Frameworks & Libraries:

Tensorflow, Pytorch, fastai, Qdrant, Scikit-Learn, OpenCV, RASA, NLTK, Spacy, Flask, FastAPI, Pandas, Numpy, Matplotlib, etc.

Applications & Platforms:

Linux, Git, Hugging Face, AWS Sagemaker, EC2, S3, CloudWatch, AWS Batch, Lambda, Docker, VS-Code, PyCharm, IPython Notebooks, etc

Research

- "Deep Learning for Bias Detection: From Inception to Deployment", (December, 2021), AusDM, Australia.
- "Imagenics Super-Resolution Generative Adversarial Networks" (May, 2020), IJCSE.

Certificates

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- Python for Data Science Professional, Edureka.
- Natural Language Processing with Classification and Vector Spaces, Coursera.
- Introduction to Machine Learning in Production, Coursera.
- AWS Technical Professional, AWS.

Work Experience

Data Scientist

Credgenics, Noida, India | August 2022 - Present

- Project:** Built RAG on legal notices & financial law documents relevant for borrowers.
- Solution:**
 - Used open source LLMs, Qdrant vector DB, OpenAI APIs, searching & Ranking.
 - Worked on micro service development using FastAPI & RAG deployment.
- Project:** In-house Chatbot & Voicebot development for loan collection use case.
- Solution:**
 - Worked on RASA, built own framework for visual chatbot builder, Speech-to-Text, Text-to-Speech services and their deployment.
 - Built classification models using BERT, DistilBERT, worked on knowledge distillation and model optimisation using Quantization and ONNX runtime.
- Project:** Built In-house call centre auditing with real-time agent assist & automated QA capable of processing more than 1 million calls in a month.
- Solution:**
 - Worked on open source STT models like Whisper & performed speaker diarization.
 - Used LLMs to process transcription & built agent assistance & QA frameworks.
- Project:** Built Risk Segmentation models in loan collection use case for Banks/NBFCs.
- Solution:**
 - Created ML classification models using XGBoost, worked on model evaluation using AUC, Decile statistics and Gains table.

Software Engineer (Data Science)

To The New, Noida, India | February 2020 - August 2022

- Project:** Detection of unconscious bias in enterprise content like Job Descriptions, social media posts, career blogs, candidate emails, etc.
- Solution:**
 - Built classification models using LSTMs, DistilBERT to classify text into Gender, Age, and Disability bias, etc.
 - Finetuned T5 model for generating unbiased text recommendations.
 - Worked on API development and model deployment.
- Project:** Detection of toxicity, microaggressions, stereotypes, etc in enterprise conversational content and integration with platforms like Slack.
- Solution:**
 - Worked on transformer-based BERT, RoBERTa, DistilBERT models to identify issues in text.
 - Deployed the models using Flask and Docker containers and built a Slackbot integration as a SaaS product to alert users if the message contains any issue.

Academic History

B.Tech. - Computer Science & Engineering.

IMS Engineering College AKTU, India | (2016-2020)

- Cumulative GPA: 8.82/10
- Participant and winner of various tech competitions at IIT Hyderabad, IIT Kanpur, and IIT Bombay.

Contact

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